

Early Detection of Alzheimer's Disease Using Machine Learning

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Abstract: Alzheimer's disease (AD) develops through a long pre-symptomatic phase in which neurobiological damage builds silently for up to two decades before any clinical sign appears. By the time a patient shows noticeable symptoms, the window for effective treatment has already narrowed. This paper describes a machine learning pipeline built for early AD progression detection, trained on longitudinal data from the Alzheimer's Disease Neuroimaging Initiative (ADNI). We combine data from ten clinical assessment tables covering demographic, genetic (APOE4), neuropsychological, functional, and structural brain imaging information into a single 19-feature matrix. Five classifiers (Logistic Regression, Support Vector Machine, XGBoost, Random Forest, and a Multi-Layer Perceptron) are trained under 5-fold stratified cross-validation and compared side by side. All models reach a mean ROC-AUC above 0.95, with XGBoost achieving the highest recall (0.961) and Logistic Regression the highest F1-score (0.776). The final system is deployed as a prediction tool intended to assist clinicians in identifying high-risk patients for earlier referral, not to substitute clinical judgment.

Index Terms: Alzheimer's disease, early detection, machine learning, ADNI, MCI conversion, longitudinal biomarkers, multimodal data fusion, classification.

I. INTRODUCTION

A. Background and Motivation

Alzheimer's disease (AD) is the leading cause of dementia worldwide, responsible for 60–70% of all dementia cases. The World Health Organization estimates that around 55 million people are currently living with dementia, and that number is expected to reach 139 million by 2050 [2]. Despite years of research, there is still no cure, and the disease progresses in a way that cannot be reversed once symptoms appear.

One of the most underappreciated aspects of AD is how early the damage begins. Changes in the brain, such as amyloid-beta plaque buildup, tau tangle formation, and shrinkage of the hippocampus, start 15 to 20 years before a patient notices any memory problems [9]. By the time someone reaches a neurologist with clinical complaints, a large portion of brain damage is already done, and the opportunity to step in early has largely passed.

The financial burden adds to the human cost. Global dementia care is estimated to cost over \$1.3 trillion per year, with the bulk of spending concentrated in late-stage care. Catching patients earlier could move a significant share of that care to a stage where it is both more effective and less expensive.

B. Problem Statement

In current clinical practice, a diagnosis typically comes only after a patient or family member reports memory problems, at which point the disease is already well advanced. Standard screening tools like the Mini-Mental State Examination (MMSE) only pick up problems after meaningful cognitive decline has already taken place. This wait-and-see approach limits what any treatment can actually do.

What is missing is a proactive screening tool that can pick up on subtle warning signals across cognitive tests, genetic markers, and brain scans at the same time. No single clinician can process all of those signals simultaneously, but machine learning can.

C. Proposed Approach

This paper proposes a machine learning system trained on data from the Alzheimer's Disease Neuroimaging Initiative (ADNI), the largest publicly available long-term study of AD biomarkers in the world [1]. By bringing together demographic data, APOE4 genetic risk markers, neuropsychological test scores, and structural brain imaging measurements, the system learns to recognise the pattern of features that predicts AD progression, often years before a clinical diagnosis would be made.

Five machine learning models (Logistic Regression, SVM, XGBoost, Random Forest, and a Neural Network) are trained, tuned, and compared. The result is a decision support tool designed to help clinicians flag high-risk patients earlier, so that referrals, further tests, and care planning can begin sooner.

D. Clinical and Social Impact

Finding at-risk patients early has real benefits for everyone involved. Patients and their families can plan ahead while the patient can still take part in decisions about their own care. Doctors can recommend lifestyle changes like exercise, cognitive training, and diet adjustments, all of which have shown some evidence of slowing the disease at early stages [9]. Patients can also be enrolled in clinical trials at the point where experimental treatments have the best chance of working. For healthcare systems as a whole, planned early care is far less costly than crisis-driven late-stage management.

II. RELATED WORK

A. ADNI as the Standard Benchmark

Since its launch in 2003, ADNI has become the go-to dataset for computational AD research [1]. It brings together MRI and PET brain scans, genetic profiles, cognitive test scores, and cerebrospinal fluid (CSF) biomarkers, all collected from the same group of participants over many years. A thorough 2024 review confirms that this combination of modalities and longitudinal design makes ADNI uniquely suited to early AD

biomarker research, and it serves as the reference point for nearly all modern ML work in this area [9].

B. Classical Machine Learning for MCI Conversion

The most relevant prior studies focus on using structured ADNI data to predict which patients with Mild Cognitive Impairment (MCI) will go on to develop AD. A 2024 study published in the *Journal of Medical and Biological Engineering* tested eight different classifiers on 819 ADNI patients [3]. XGBoost came out on top, striking the best balance between accuracy and sensitivity under imbalanced class conditions. This finding was one of the key motivations for including XGBoost in our pipeline.

A bioRxiv preprint that applied Random Forests to a clinical ADNI sample ($n = 383$) reported an accuracy of 93.6% and a ROC-AUC of 96% for predicting MCI-to-AD conversion, beating earlier SVM-based work that reached AUC values around 88% [11] and 84.6% [12]. Logistic Regression on the same feature set reached about 0.90 AUC, giving us a useful baseline [4].

C. Multimodal Data Fusion

Study after study has shown that combining different types of data beats any single source on its own. A *Nature Scientific Reports* paper fused MRI images processed by a 3D Convolutional Neural Network with electronic health records and genomic SNP data from ADNI, and the multimodal model outperformed SVM, Random Forest, and k -NN baselines [5]. A 2025 study in *Alzheimer's & Dementia* built a hybrid framework that paired an LSTM for tracking biomarker changes over time with a feedforward network for static features, again beating single-modality models [8]. Our work follows this same logic by bringing five different data domains together into one feature matrix.

D. Longitudinal vs. Cross-Sectional Modelling

A 2024 *iScience* study used machine learning on multi-year ADNI data to build a composite index of disease progression that predicted time-to-AD-conversion with a C-Index of 0.923, well above what hippocampal volume alone (0.774) or MMSE alone (0.808) could achieve [6]. A separate PMC study showed that a longitudinal LSTM reached AUC 0.93, beating a cross-sectional Random Forest at 0.90 for the same 2-year prediction task [7]. These findings underline how much information is carried in how a patient changes over time, not just where they stand at any single visit.

E. Key Predictive Biomarkers

The literature points to a consistent set of features that matter most for predicting AD: shrinkage of the hippocampus, rising CDRSB and ADAS-Cog scores, falling MMSE scores, APOE4 genotype (two copies raise risk by 8 to 12 times compared to non-carriers), and poor delayed recall on memory tests. A 2025 MDPI *Diagnostics* paper using SHAP explainability confirmed that APOE4 count, CDRSB, and hippocampal volume consistently rank among the most important features across models [10]. Our 19-feature set was built to cover all of these known predictors.

F. Class Imbalance in AD Datasets

One challenge that comes up in every AD progression study is that stable patients outnumber progressors by a wide margin, because most MCI patients do not convert to AD within any given follow-up window. The literature offers a few ways to

handle this: NEAR-MISS undersampling, class-weighted losses, and threshold shifting [3].

Our approach uses class-weighted loss functions (`class_weight='balanced'`), which give more weight to minority-class samples during training without generating synthetic data. This method is efficient and performs on par with oversampling on structured tabular data. On top of that, the SVM uses a lower decision threshold (0.25) to make the model more sensitive to progressors, boosting recall without changing the underlying probability outputs.

III. DATASET

A. Data Source

All data were obtained from the ADNI database at adni.loni.usc.edu [1]. ADNI is a long-running multisite study that began in 2003 and is funded through a public-private partnership. Its core goal is to identify and validate biomarkers for tracking MCI and early AD over time using repeated MRI, PET, clinical, and neuropsychological assessments. The study spans four phases: ADNI1, ADNI-GO, ADNI2, ADNI3, and ADNI4.

B. Extracted Data Tables

Ten CSV tables were downloaded from the ADNI portal, each covering a different area of clinical assessment. Table 1 gives a summary of each table and its role in the pipeline.

Table 1: ADNI tables and clinical roles.

Table	Variables & Role
DXSUM	DIAGNOSIS, EXAMDATE. Primary diagnosis (CN=1, MCI=2, Dementia=3); used to build the PROGRESSOR label.
PTDEMOG	PTGENDER, PTEDUCAT, PTMARRY, PTDOB. Demographic risk factors; education level is a proxy for cognitive reserve.
APOERES	GENOTYPE converted to APOE4_COUNT. Allele count (0/1/2); the strongest known genetic risk factor for late-onset AD.
MMSE	MMSCORE. Global cognitive screening on a 0 to 30 scale.
CDR	CDRSB. A staging tool sensitive across all levels of AD severity.
ADAS	TOTSCORE, TOTAL13. The primary cognitive outcome measure used in AD clinical trials.
FAQ	FAQTOTAL. Measures functional independence in everyday tasks.
NEUROBAT	LIMMTOTAL, LDELTOTAL, AVTOT1-6, AVDEL30MIN, CATANIMSC. Covers verbal memory and language fluency.
FOXLABBSI	BRAINVOL, VENTVOL. Whole brain and ventricular volumes from MRI.
UCD-WMH	TOTAL_HIPPO. Hippocampal volume; the most widely used structural biomarker for early AD.

C. Target Variable Construction

The binary label, PROGRESSOR, was built from the DXSUM table across all follow-up visits. A patient is labelled PROGRESSOR = 1 if their diagnosis reaches Dementia/AD at any visit after baseline, and PROGRESSOR = 0 if they remain stable throughout the study. This approach ensures the model

is learning from the pre-symptomatic period, not just flagging patients who already have established disease.

D. Visit Code Harmonisation

A technical complication in ADNI is that each study phase uses its own visit naming scheme. ADNI1 uses month codes (m06, m12, etc.), ADNI-GO/ADNI2 uses sequential codes (v01, v03, etc.), ADNI3 uses year codes (y1, y2, etc.), and ADNI4 uses prefixed codes (4_bl, 4_m12, etc.). A custom function, `viscode_to_months()`, was written to convert all of these into a single integer MONTH variable measured from baseline, making it possible to merge records across phases consistently. Screen-fail visits and any unresolvable codes were removed.

E. Master Table Construction

The master table was built by performing sequential left-merges on the keys [RID, MONTH], starting with DXSUM as the base. Merging in this order guarantees that the diagnosis record is always present, and each new clinical domain is joined on top without creating duplicate rows. For tables that had more than one record at the same visit month (for example, NEUROBAT includes both a screening visit and a true baseline visit), a priority map was used to keep the correct baseline row. The AGE variable was derived from the difference between date of birth and examination date, expressed in fractional years.

The resulting table covers all four ADNI phases and includes thousands of participants with data from multiple visits. After deriving the target label, selecting features, and applying quality filters, the dataset moves to preprocessing.

F. Class Distribution

As expected in AD research, the PROGRESSOR label is imbalanced: stable patients outnumber those who convert to dementia across the follow-up period. This reflects the real epidemiology of the disease, where most MCI patients do not progress within a given observation window.

This imbalance matters for how we train and evaluate the models. A classifier that simply labels everyone as stable would score high accuracy but be completely useless in practice. For this reason, we focus on recall, PR-AUC, and ROC-AUC rather than raw accuracy, as these metrics are more reliable when class sizes differ.

- **Demographic/Genetic (5):** AGE, PTGENDER, PTEDUCAT, PTMARRY, APOE4_COUNT
- **Cognitive & Functional (11):** MMSCORE, CDRSB, FAQTOTAL, TOTSCORE, TOTAL13, LIMMTOTAL, LDELTOTAL, AVDEL30MIN, AVDELTOT, CATANIMSC, RAVLT_AVERAGE
- **Brain Imaging/MRI (3):** BRAINVOL, VENTVOL, TOTAL_HIPPO

IV. METHODOLOGY

A. Pipeline Overview

The pipeline runs in sequence as follows: (1) loading raw ADNI CSV files and selecting relevant columns; (2) converting visit codes to a unified MONTH variable; (3) merging all tables into a longitudinal master table; (4) deriving the PROGRESSOR label from DXSUM; (5) encoding APOE4 allele counts from raw genotype strings; (6) imputing missing values using MICE; (7) engineering the RAVLT_AVERAGE feature; (8) configuring class weights to handle imbalance; (9) running 5-fold stratified

cross-validation with fold-level scaling; (10) training and tuning each model; and (11) exporting the trained models for the prediction interface.

Every preprocessing step is applied within the training fold only, so no information from the validation set leaks into model training. The cleaned output file (`df_clean_AD.csv`) is shared across all five models to keep comparisons fair.

B. Data Extraction and Merging

All ten raw ADNI CSV files were loaded and checked against the ADNI Data Dictionary to confirm the clinical meaning of each variable. A schema was then defined to specify exactly which columns to keep from each table, based on what the AD literature identifies as meaningful predictors.

The master table was built through sequential left-merges on [RID, MONTH], with DXSUM as the starting point. Demographics, genetic data, ADAS, FAQ, brain volumes, MMSE, CDR, neuropsychological scores, and hippocampal volume were all joined in order. Where a table had multiple rows for the same month (such as NEUROBAT with both screening and baseline records), a priority map was applied to keep the correct baseline. AGE was calculated from the difference between date of birth and examination date.

C. Exploratory Data Analysis

The PROGRESSOR class split was visualised using count plots and pie charts. Diagnosis breakdowns by cohort and outcome group were produced to get a clear picture of the patient population before modelling.

Distribution plots were generated for all 22 candidate features, organised by domain: bar charts for categorical demographics, histograms with kernel density overlays for cognitive scores (including skewness annotations), and volume histograms for MRI features. Bivariate associations between each feature and the PROGRESSOR label were tested using Mann-Whitney U tests for continuous variables and point-biserial correlation, confirming that the chosen features show statistically significant differences between groups.

D. Preprocessing Pipeline

APOE4 Encoding

The GENOTYPE field in APOERES contains text strings such as 2/3, 3/4, and 4/4. These were parsed to extract the count of APOE4 alleles (0, 1, or 2), clipped to the valid range, and stored as an integer.

Missing Value Imputation

Brain imaging variables in particular have a high rate of missing data across ADNI tables. To handle this, Multiple Imputation by Chained Equations (MICE) was applied using `scikit-learn`'s `IterativeImputer` backed by an `ExtraTreesRegressor`. This method fills in each missing value using all other features as predictors, which preserves the natural relationships between variables. Simple mean or median imputation was avoided because it compresses the spread of values and distorts variable distributions.

Feature Engineering

The six RAVLT learning trial scores (AVTOT1 through AVTOT6) were averaged into a single feature called

RAVLT_AVERAGE. This captures the overall verbal learning rate in one number rather than six correlated columns, reducing redundancy without losing the core information from the learning trajectory.

Class Imbalance Handling

To address the imbalance between progressors and stable patients, class-weighted loss functions (`class_weight='balanced'`) were used in all applicable classifiers. This gives more training weight to the minority class in proportion to how underrepresented it is, without generating any synthetic patients. The SVM additionally uses a lower decision threshold (discussed below) to further improve sensitivity.

E. Machine Learning Models

All five models were evaluated under 5-fold stratified cross-validation (`StratifiedKFold`, $k = 5$). A `StandardScaler` was fitted on each training fold and applied to the corresponding validation fold, ensuring that scaling parameters from validation data never influence training.

Logistic Regression

L1 (LASSO) regularisation was used via the SAGA solver, with $C = 0.577$, `class_weight='balanced'`, and `max_iter=1000`. L1 regularisation pushes the coefficients of weak predictors to exactly zero, effectively selecting only the most informative features. This makes the model easy to interpret: a clinician can look at the non-zero coefficients and immediately see which variables are driving a high-risk prediction.

Support Vector Machine

A linear kernel SVM with Platt-scaled probability outputs was used. The decision threshold was set to 0.25 rather than the standard 0.50. This was a deliberate clinical choice: in an AD screening context, missing a patient who will develop the disease is a much more serious error than a false alarm, which only results in a follow-up appointment. Lowering the threshold shifts the balance toward sensitivity, at a modest cost to precision.

XGBoost

The XGBoost classifier used the `gbtree` booster with a binary logistic objective, optimised for AUC. Class imbalance was handled through `scale_pos_weight`, set to the ratio of negative to positive training samples in each fold. Hyperparameters were selected via `RandomizedSearchCV`, yielding the following final configuration: 450 estimators, a learning rate of 0.012, maximum tree depth of 5, minimum child weight of 14, subsample ratio of 0.78, column subsampling ratio of 0.80, and regularisation terms $\gamma = 0.36$, $\alpha = 1.36$, and $\lambda = 1.95$. The decision threshold was kept at the default 0.50. At each fold, diagnostics recorded included ROC-AUC, PR-AUC, F1-score, accuracy, recall, precision, and the train-versus-validation AUC gap as a check for overfitting. The gradient-boosted tree structure is well-suited to structured tabular data, as confirmed across multiple prior AD studies [3].

Random Forest

The tuned hyperparameters included `n_estimators`, `max_depth`, and `min_samples_split`, selected through grid search. Beyond classification, the Random Forest produces feature importance scores based on mean decrease in impurity. Based on prior literature, we expected hippocampal volume, CDRSB, MMSCORE, LDELTTOTAL, and APOE4_COUNT to rank highest [4, 10].

Multi-Layer Perceptron (MLP)

The MLP was built using the Keras Sequential API, with multiple fully connected layers, Dropout regularisation between layers, and L2 weight regularisation to prevent overfitting. Class weights were passed into the training loss. Two callbacks were used: `EarlyStopping` (patience = 10, restoring the best weights) to stop training when the validation loss stopped improving, and `ReduceLROnPlateau` to drop the learning rate when progress stalled. The trained model was saved in Keras format; the scaler and threshold were pickled separately.

F. Model Export and Deployment

All five trained models were saved for deployment. A web-based prediction interface was built so that a clinician can enter a patient's 19 values and receive predictions from all five models at once. The interface displays the predicted probability of progression and the binary classification decision from each model side by side, so the clinician can see whether the models agree and how confidently. Feature labels are written in plain clinical language rather than technical variable names.

G. Evaluation Framework

In addition to the standard cross-validation metrics, several further diagnostics were used. A train-versus-validation AUC gap was tracked at each fold to check for overfitting. A large gap would suggest the model is memorising training data rather than learning general patterns.

Precision-Recall curves were plotted for all five models. Unlike ROC curves, PR curves focus on the minority class and give a clearer picture of how well the model handles progressors specifically, which matters most in a clinical screening scenario.

A confusion matrix was recorded at each fold, and the mean true positive rate (recall), false positive rate, and false negative rate were calculated across folds. The false negative rate is particularly important here, as it tells us directly what fraction of future AD patients would be missed.

Probability calibration was also assessed. A well-calibrated model should produce probabilities that match observed event rates: a patient with a predicted probability of 0.7 should, across a large group, convert to AD about 70% of the time. The MLP's `EarlyStopping` callback and the SVM's Platt scaling both help with calibration.

V. RESULTS AND DISCUSSION

A. Model Performance

All five models were trained and tested under the same conditions. Table 2 shows the mean performance metrics across the five cross-validation folds.

Table 2: Mean cross-validated performance across all classifiers.

Model		ROC-AUC	F1	PR-AUC	Recall
Log. Regres-	sion	0.954	0.776	0.870	0.897
SVM		0.954	0.766	0.865	0.907
XGBoost		0.954	0.728	0.870	0.961
Random Forest		0.954	0.732	0.871	0.942
MLP		0.952	0.736	0.862	0.935

B. Overall Performance Analysis

All five models land at roughly the same ROC-AUC of 0.954, which tells us that the 19-feature multimodal representation is carrying most of the predictive weight, regardless of which model is used. This lines up with prior work showing that well-built feature sets on structured ADNI data tend to produce AUC values in the 0.90 to 0.96 range across many different classifiers [3, 4].

The fact that all models perform so closely also points to the preprocessing pipeline as a key contributor. The MICE imputation preserving inter-feature relationships, the harmonised visit coding, and the RAVLT_AVERAGE aggregation all appear to have lifted the floor for every model.

It is worth noting the gap between ROC-AUC (around 0.954) and PR-AUC (0.862 to 0.871). ROC-AUC treats both classes equally, while PR-AUC zeroes in on the minority class. The lower PR-AUC reflects how hard it is to pinpoint progressors with precision in an imbalanced setting, and reinforces the importance of threshold tuning for real clinical use.

C. Model-Specific Analysis

XGBoost

XGBoost reaches the highest recall at 0.961, consistent with the 2024 *Journal of Medical and Biological Engineering* findings and the bioRxiv Random Forest study, both of which show that ensemble tree models tend to outperform linear and kernel-based approaches on structured clinical data [3, 4]. For a screening tool where catching every true progressor is the main priority, XGBoost is the strongest option.

Logistic Regression

Logistic Regression, the simplest model in the set, achieves the best F1-score (0.776) and has a clear clinical advantage: its L1-regularised coefficients can be read directly as log-odds contributions. A clinician can look at the coefficient values and understand exactly which features pushed a patient into the high-risk group. This kind of transparency matters a great deal in regulated healthcare settings.

Support Vector Machine

The lower decision threshold of 0.25 raises recall to 0.907, confirming that threshold adjustment is an effective tool for managing the false-negative cost in screening contexts. Precision takes a small hit as a result, but that trade-off is acceptable when a false alarm leads only to a follow-up appointment rather than unnecessary treatment.

Random Forest

The Random Forest achieves the highest PR-AUC (0.871) and a recall of 0.942, placing it close behind XGBoost. Its additional value comes from feature importance scores, which give clinicians a ranked list of the biomarkers that drive predictions. Based on the literature, hippocampal volume, CDRSB, MMSCORE, LDELTOTAL, and APOE4_COUNT are expected to come out on top [10, 11].

Neural Network (MLP)

The MLP delivers competitive results (ROC-AUC 0.952, Recall 0.935) in spite of the fact that neural networks typically need larger datasets to beat tree-based models on tabular data. The EarlyStopping and ReduceLROnPlateau callbacks, combined with Dropout and L2 regularisation, produce a stable model that does not overfit. Its main advantage is the ability to pick up on non-linear relationships between features without any manual interaction terms.

D. Comparison with Prior Literature

Table 3 places our results alongside closely related prior work on ADNI data. Our pipeline performs at or above the current state of the art for structured clinical feature sets, while drawing on a larger and more varied feature space than most comparable studies.

Table 3: Comparison with related ADNI-based studies.

Study	Model	AUC	Feature Domains
Grassi et al. [11]	SVM	0.88	Clinical
Huang et al. [12]	RF	0.846	Clinical
bioRxiv [4]	RF	0.96	Clinical
JMBE [3]	XGBoost	0.95	Clinical
Ours	All 5	0.954	Multimodal (5)

A practical difference worth highlighting is how much data we pulled together. Most prior studies draw from two or three ADNI tables at most. Our pipeline merges ten tables through a custom harmonisation step, producing a richer picture of each patient. Offering all five trained models through a single interface also gives clinicians flexibility: XGBoost when recall matters most, Logistic Regression when an explanation is needed, and Random Forest as a middle ground.

E. Clinical Implications

The high recall values seen across all models (0.897 to 0.961) matter a great deal in practice. In AD screening, missing a patient who will go on to develop the disease is a serious failure: that person loses the chance for early intervention, lifestyle changes, and enrolment in clinical trials. A false positive, by comparison, leads only to an extra clinic visit and further testing, which is a much lighter cost.

Presenting the output as a probability rather than a yes/no decision also adds practical value. A clinician seeing a 70% progression probability can act immediately, while one seeing 35% might choose to monitor and retest. A binary classifier collapses this distinction entirely.

It is also worth pointing out that all 19 features in this model are already collected as part of routine AD assessments. MMSE,

CDR, and ADAS scores come from standard neurology consultations; APOE4 genotyping is a single blood test; and MRI brain volumes are derived from scans that are already standard of care in memory clinics. No additional or invasive procedures are required, which means this tool could realistically be added to existing clinical workflows without major disruption.

F. Limitations

A few limitations should be noted. To begin with, the ADNI cohort is drawn mainly from North America and tends toward higher education levels, so it may not generalise well to more diverse or lower-resource populations. External validation on a separate cohort (such as the UK Biobank or the EPAD dataset) would be needed to confirm that the models hold up outside of ADNI.

Second, all evaluation was done through cross-validation on a single dataset. While this is a standard and accepted approach, it does not replace the evidence that would come from testing on a genuinely independent patient sample.

Third, the feature set is mostly based on single time-point measurements. Longitudinal features, such as how fast MMSE or hippocampal volume changes between visits, are likely to add predictive power, as shown in LSTM-based work in the literature [6, 7].

Fourth, the uncertainty introduced by MICE imputation is not carried forward into the model predictions. A Bayesian or uncertainty-aware framework could report confidence intervals alongside each prediction, which would give clinicians a better sense of how much to trust any given result.

Finally, the pipeline does not yet use PET-derived amyloid or tau biomarkers. These are the earliest detectable biological signs of AD pathology, and adding them is expected to push early detection performance further as PET becomes more widely available in clinical settings.

VI. CONCLUSION

This paper describes a full machine learning pipeline for detecting Alzheimer’s disease progression at an early stage. ADNI data from ten clinical tables was harmonised across four study phases, merged into a single longitudinal table, imputed using MICE, and processed into a 19-variable multimodal feature set. Five classifiers of different complexity were then trained and tested under rigorous cross-validation, and all reached a ROC-AUC above 0.95 with recall above 0.89.

The main takeaway is that a well-built feature set from routine clinical data is enough to reach near state-of-the-art performance for AD prediction, without needing specialised imaging beyond standard MRI. The fact that five very different models all perform similarly suggests that the preprocessing and feature engineering are doing a lot of the heavy lifting.

From a practical standpoint, the three most useful outputs of this work are: (1) XGBoost’s recall of 0.961, meaning it would flag about 96 out of every 100 future AD patients for follow-up; (2) the interpretable coefficients of the Logistic Regression model, which can be shared with clinicians to explain why a patient was flagged; and (3) the unified interface that puts all five models in the hands of clinical staff without requiring any coding knowledge.

A. Future Work

There are several clear directions for extending this work. Adding longitudinal change features, such as year-over-year decline in MMSE, CDRSB, or hippocampal volume, is expected to improve predictions further, consistent with results from LSTM-based studies [6, 7]. Including PET amyloid and tau biomarkers as they become more widely collected would push the detection window earlier into the pre-symptomatic phase.

Applying SHAP explanations at the individual patient level would make the models more trustworthy and easier to adopt in regulated clinical settings. Finally, connecting the prediction interface directly to electronic health records would remove the need for manual data entry, which is currently the biggest practical barrier to using tools like this in everyday clinical care.

DATA AVAILABILITY

All data used in this study were obtained from the ADNI database, available at <https://adni.loni.usc.edu>. Access requires registration and approval through the ADNI Data Sharing and Publications Committee, and was obtained in accordance with ADNI’s data use agreement.

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